Performance Testing

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Custom benchmarking - manual response-time, CPU, and RAM profiling of the FastAPI /evaluate endpoint
Type of Testing	Load Testing (cold start, warm cache), Stress Testing (concurrent evaluations), Baseline Testing (health check)
Target Module / API	POST /evaluate - the 10-step AI inference pipeline (Whisper STT + Librosa DSP + Sentence-Transformers NLP + Scoring)
Test Environment	Cloud container - 1 vCPU, 1 GB RAM (simulating Streamlit Community Cloud), Neon PostgreSQL over HTTPS
Test Date	July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Cold Start - first evaluation after server restart; all ML models load from disk/network	1	40-55	Whisper, Bi-Encoder and Cross-Encoder models load successfully; RAM peak ~700-900 MB
2	Warm Cache - subsequent evaluation after models are cached in memory	1	15-25	Only inference runs; RAM peak ~500-700 MB, response time within the 30s NFR target
3	Topic Mismatch - audio content does not match the reference concept (short-circuit path)	1	8-12	Topic Guardrail triggers; pipeline exits early after Whisper + Guardrail, skipping SBERT/Cross-Encoder
4	Concurrent Evaluations - two simultaneous /evaluate requests (stress test)	2	25-50 each	Requests serialised by the GIL; 0% error rate with models already cached (~800-950 MB RAM)

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~15-25s (warm cache)	Fail (vs. template target)	Meets the project's own NFR-01 target of <= 30s for a 60s audio clip
2	Response Time (Max)	< 5 seconds	~40-55s (cold start)	Fail (vs. template target)	Within the relaxed cold-start allowance of <= 60s while models load
3	Throughput (Req/sec)	-	1 concurrent request	Marginal	Single-threaded (GIL); Celery + Redis async workers planned to raise throughput
4	Error Rate	< 1%	0%	Pass	No failures observed across cold start, warm cache, topic mismatch, and concurrent scenarios
5	CPU Utilization	< 80%	60-90% (inference bursts)	Marginal	Bursts occur during Whisper and Sentence-Transformers inference, then drop after
6	Memory Utilization	< 80%	~50-70% warm / up to ~90% cold	Marginal	Whisper tiny model + gc.collect() keep peak RAM close to the 1 GB Streamlit Cloud ceiling

Step 4: Observations & Analysis

Key Findings: The warm-cache pipeline completes in ~15-25 seconds, within the project's own NFR-01 target of <=30 seconds for a 60-second clip. Cold start (first request after deployment) takes 40-55 seconds due to model loading, which is expected and acceptable given the relaxed cold-start allowance. Bottlenecks Identified: Whisper transcription (5-12s) and the combined Bi-Encoder + Cross-Encoder NLP inference (4-10s) are the largest contributors to pipeline latency. The synchronous, single-threaded execution model also limits the system to one evaluation at a time. Optimization Steps Taken: gc.collect() is called after Whisper and Librosa operations to free intermediate tensors; all three ML models use module-level singleton caching to avoid reloading on every request; Whisper's tiny (39M parameter) model is used instead of larger variants; PDF reports are generated entirely in-memory (io.BytesIO) to avoid disk I/O overhead.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Response-time and RAM-usage benchmark logs for the Cold Start, Warm Cache, Topic Mismatch, and Concurrent scenarios to be attached here]